

JIAYU ZHANG

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Education

Columbia University, Ph.D. in Operations Research, Aug. 2024- **Present**

Columbia University, M.S. in Operations Research, Aug. 2024 - Oct. 2025

Shanghai Jiao Tong University, B.S. in Mathematics, Aug. 2020 - Jun. 2024

Publications

Preprints

- Jiayu Zhang and Tianyi Lin. "Scale-Invariant Neural Network Optimization: Norm Geometry and Heavy-Tailed Noise." arXiv preprint arXiv:2605.18528 (2026).
- Donald Goldfarb, Lexiao Lai, Tianyi Lin, and Jiayu Zhang. "Non-Convex Self-Concordant Functions: Practical Algorithms and Complexity Analysis." arXiv preprint arXiv:2511.15019 (2025).
- Mingyu Lei, Jiayu Zhang, and Yinyu Ye. "Blessing of high-order dimensionality: from non-convex to convex optimization for sensor network localization." arXiv preprint arXiv:2308.02278 (2023).

Undergraduate Thesis

- Jiayu Zhang. "Accelerated Gradient Descent with Long Step-Size." Shanghai Jiao Tong University (2024). Supervisor: Jingwei Liang.

Experiences

- Cardinal Operations, Student Researcher, Feb. 2023 - Sep. 2023, Shanghai, China

Teaching

- Spring 25: IEOR 4004 Opt. Models and Methods, Prof. Vineet Goyal, Columbia IEOR.
- Fall 25: IEOR 6613 Opt. I (Ph.D.), Prof. Vineet Goyal, Columbia IEOR.
- Spring 26: EEOR 6616 Convex Opt. (Ph.D.), Prof. Bento Natura, Columbia IEOR.

Fellowships and Awards

- Fall 2024: Boyle Fellowship, Columbia University.

Skills

- Coding: Python, Julia, MATLAB, R; PyTorch, JuMP; Linux, Git, Bash, Slurm, LaTeX.
- Languages: English (proficient), Mandarin (native).